1. SOFTWARE PROJECT MANAGEMENT (PhDCS 103)

MODULE I

Overview of Project Management, PMI Processes, Software project phases, Organizational structures, Project charter, Statement of Work (SOW) **Planning Phase,** Development lifecycle models, Matching lifecycles to projects, Project plans, Work Breakdown Structures (WBS) **Estimation and Budgeting**, Estimation, Budgeting, Project, selection, NPV, ROI, Payback models

MODULE II

Scheduling, Project network diagram fundamentals, PERT techniques, Gantt charts, Critical chain scheduling **Risk and Change Management**, Mid-term review , Risk management, Change control, More MS-Project.

Development Management, Team models, Requirements process, Configuration, management, Software metrics, Programming languages & tools, Managing conflict and motivating, MS-Project: Assigning Resources.

MODULE III

Project Control, Status reporting, Project metrics, Earned value analysis, Communications Techniques, Process Improvement, MS Project: (a) Resource leveling (b) Other views

System Test Process, Test specifications, Black box and white box testing, Test scripts, Unit and integration testing, Acceptance test specifications, Test tools, MS Project:(a) Reporting.

Final Phases & Other Issues, Project Recovery, Documentation, Cutover/Migration Post Project Reviews, Closing, MS Project: (a) Advanced features.

Reference Books:

1. S. McConnell, "Software Project Survival Guide" (1997)

2. S. Berkun, "The Art of Project Management", (2005)

3. C. Larman, "Agile and Iterative Development: A Manager's Guide", (2003)

4. W. Royce, "Software Project Management: A Unified Framework", (1998)

5. J. Highsmith, "Agile Project Management: Creating Innovative Products", (2004)

6. T. DeMarco, "The Deadline: A Novel About Project Management", (1997)

7. T. DeMarco, "Peopleware: Productive Projects and Teams", (1999)

8. E. Bennatan, "On Time Within Budget: Software Project Management Practices and Techniques", (2000)

2. NETWORKING PROTOCOLS (PhDCS 104)

MODULE I

Networks and Services, Approaches to Network Design, The OSI Reference Model; Overview of TCP/IP Architecture, Application Protocols and TCP/IP Utilities, Internet Architecture Interconnection through IP Routers, Internet Protocol (IP), User datagram protocol (UDP).

MODULE II

Routing Cores - peers Algorithms Autonomous Systems Exterior Gateway Protocol Multicast Address. Internet Group Management Protocol (IGMP) and Implementation. TCP/IP over ATM networks: ATM cell Transport, Adaptation Layer, IP Address Building in an ATM network Logical IP subnet Concept ATMARP packet format. Domain name system**,** Remote Login (Telnet, Rlogin) File Transfer and Access (FTP, TFTP, NFS), Electronic mail (SMTP, MIME) Internet Management (SNMP, SNMPV2) Internet Security and Firewall Design Post Office Protocol (POP) Network News Transfer Protocol (NNTP).

MODULE III

TCP/IP over view- The Transport Layer: TCP and UDP. Elementary TCP Sockets. TCP Client-Server Example. I/O Multiplexing: The select and poll Functions. Socket Options. Elementary UDP Sockets. Elementary Name and Address Conversions. The Client Server Model and Software Design, Concurrent Processing in Client-Server Software, Iterative, Connectionless Servers (UDP), Iterative, Connection-Oriented Servers (TCP), Concurrent, Connection-Oriented Servers (TCP). Single-Process, Concurrent Servers (TCP). Multiprotocol Servers (TCP, UDP), Multiservice Servers (TCP, UDP). Uniform, Efficient Management of server. Concurrency in clients. TCP/IP Architecture, The Internet Protocol, Limitations of IPv4 and Introduction to IPv6, User Datagram Protocol, Transmission Control Protocol, DHCP, Introduction to Internet Routing Protocols

Reference Books:

1. A. Leon-Garcia, Indra Widjaja, "Communication Networks", Tata McGraw Hill, 2000

2. William Stallings, "Data and Computer Communications", Pearson Education, 7th Edition.

3. Andrew S. Tanenbaum, "Computer Networks", Prentice Hall India, 4th Edition, 2003

4. W.Richard Stevens: TCP/IP Illustrated vol 1: The Protocols, Pearson Edun. Asia, 2000.

5. Douglas Comer: Internetworking with TCP/IP vol.1: Principles, Protocols and Architecture, Prentice Hall, 4th edition, 2000

3. REAL TIME SYSTES (PhDCS 105)

Module 1